Do you need an AI PC?

If you just use online AI chatbots or run AI apps off the cloud, you DO NOT need an AI PC. A normal PC with good specs and a stable, high-speed internet connection is all you need.

You need an AI PC if you run AI apps or models locally on your computer, for varied reasons like customisation, automation, speed, performance, efficiency, or privacy. For example, if

- You are an AI app developer,
- You are building an AI agent,
- You need to customise or train an AI model locally,
- You work with sensitive data or large data sets and do not trust the cloud,
- You are a content creator who uses AI for editing, and image rendering,
- Your use-case will not tolerate latency—like live translation, 3D rendering, webcam image enhancement or noise cancellation,
- Or, if you do a lot of computational modelling.

Once you decide you need an AI PC, you will have to make a choice between a laptop or a desktop. If portability is your primary concern, a laptop is the obvious choice. But, if you are looking for more power, then desktops/workstations might be preferable, as they are more scalable and have better cooling systems as well.

parallel processing, like analysing large amounts of data, rendering images, editing videos, training AI models, and so on. They are commonly found in enterprise and gaming PCs.

With the AI boom, the logical progression was to use the same GPUs for developing and running AI models locally on PCs. But, while GPUs are good for training AI models, engineers soon realised they are not really cut out for repetitive AI inference. As GPUs are power-hungry creatures, using them for on-device AI/ML inference tends to drain the battery and heat up the device. This leads to power and thermal throttling, that is, a reduction in clock speeds and a slowing down of the GPU to protect the hardware. Plus, repetitive inference tends to hog the GPUs, leaving them unavailable for tasks they are originally meant to do!

This dilemma led to the development of neural processing units or NPUs—chips specialising in repetitive AI/ML tasks—to work alongside CPUs and GPUs in AI devices. NPUs have massive parallel processing abilities, but are smaller, less expensive, have a low latency, and a much greater power efficiency than GPUs. The development of NPUs is a critical milestone in AI history, as it has given edge AI a much-needed boost.

More about NPUs - The catalyst of AI in edge devices

NPUs are specialised hardware accelerators for efficient, real-time AI inference on edge devices like laptops and smartphones. They are purpose-built for neural networks, with an architecture inspired by aspects of the human brain. NPUs use interconnected nodes (artificial neurons) and layers, similar to the brain's structure, to process information. They can perform many calculations simultaneously (trillions per second), mirroring the brain's ability to handle vast amounts of data in parallel. They are optimised for matrix operations commonly used in AI and ML and are characterised by their low latency and remarkable power efficiency.

Unlike GPUs, which are general purpose processors adapted for AI training and inference, NPUs have been built from the ground up to handle neural network operations.

An NPU consists of thousands of specialised hardware units optimised for operations like matrix multiplications, convolutions, and activations that power deep learning models. These small processing units are capable of massive parallelism and can work on different parts of an AI model or dataset simultaneously, enabling rapid, concurrent execution of complex algorithms.

As AI inference does not require precision, NPUs generally use lower-precision numbers (8-bit or lower) instead of high-precision floating-point numbers used by CPUs and GPUs, to reduce computational complexity and thereby increase energy efficiency and speed, while reducing the memory footprint. To prevent delays caused by moving data between external memory and the processor, most NPUs include on-chip static random access memories (SRAMs) like caches and scratchpad memory.

The arrangement of the processing elements and memory, and the optimisation of data and workflow, are critical to improving the performance of an NPU. The performance of an NPU is measured in tera/trillion operations per second (TOPS). Modern NPUs boast of 35-50 TOPS. Some manufacturers mention the total system TOPS, which includes the performance of the CPU and GPU, while others rate the NPU capability alone. When comparing options, make sure you are looking at the same measure.

Working in tandem

While large data centres may have standalone NPUs, in personal devices and other edge devices, the NPU is combined with other coprocessors, such as the CPU and GPU, into a single system-on-chip (SoC).

An AI PC's firmware and customised operating system (OS) will ensure that each processor does what it is cut out for, freeing up the others for their work. Since a lot of sensitive data